

IT-DBS1 Exam S2026

Before the exam

Run both:

- Simple_Goodreads_ddl.sql
- Simple_Goodreads_data.sql

in DataGrip, so that you have the database available when you show up for the exam.

These files will be available in Itslearning and WISEflow at the latest one week before the exam.

Also, it's recommended to have Astah installed, and working, when you show up for the exam.

Exam information

The exam is 20 minutes for each student, including grading and feedback.

You will draw one question with two parts:

- The first part focuses on one of your course assignments.
- The second part will require you to either write and explain a given SQL query using the Simple_Goodreads database, or create an ER diagram.
You will get a small case at the exam for the second part.

考试前

同时运行两者：

- Simple_Goodreads_ddl.sql
- Simple_Goodreads_data.sql

在 DataGrip 中，以便你在参加考试时能够使用该数据库。

这些文件最迟将在考试前一周通过Itslearning和WISEflow提供。

此外，建议你在参加考试时已安装并能正常使用 Astah。

考试信息

考试是 每位学生20分钟，包括评分和反馈。

你将抽取一道包含两个部分的问题：

第一部分重点介绍你的一门课程作业。

第二部分要求你使用Simple_Goodreads数据库编写并解释一个给定的SQL查询，或者创建一个ER图。

考试第二部分时，你会得到一个小盒子。

Question 1 – SQL Queries

Tasks: 1. 提交你的**作业1**，并说明你是如何使用连接、分组、聚合函数和集合操作的。解释一下你从查询结果中获得的一些见解。

1. Present your **Assignment 1** and explain how you used joins, groupings, aggregate functions, and set operations. Explain some of the insights you got from the results of the queries.
2. Based on a small given case, draw an ER diagram for the situation described in the case.
2. 根据一个给定的小型案例，绘制出描述该案例情况的实体关系图。

Question 2 – Entity-Relationship Modelling

Tasks: 1. 提交你的**作业2**，重点说明你是如何发现实体和关系的，以及如何设计该图表的。

1. Present your **Assignment 2**, focusing on how you found the entities and relationships, and designed the diagram.
2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.
2. 根据一个具体的小案例，编写并解释一条SQL查询语句，以提供该案例所需的答案。

Question 3 – Enhanced ER Modelling

Tasks: 1. 提交你的**作业2**，重点说明你如何添加了EER功能以及为何需要这些功能。

1. Present your **Assignment 2**, focusing mainly on how you added EER features and why they were needed.
2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.
2. 根据一个具体的小案例，编写并解释一条SQL查询语句，以提供该案例所需的答案。

Question 4 – Mapping EER Models to Relational Schemas

Tasks: 1. 提交你的**作业3**，重点说明你是如何将EER图映射为关系模式的，以及你在主键和外键方面的考虑。

1. Present your **Assignment 3**, focusing on how you mapped your EER diagram into a relational schema, and your considerations about primary keys and foreign keys.
2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.
2. 根据一个具体的小案例，编写并解释一条SQL查询语句，以提供该案例所需的答案。

Question 5 – Global Relations diagrams

Tasks:

1. 提交你的作业3，重点展示你的GR图，并说明它与之前映射的ER图有何不同。
1. Present your **Assignment 3**, focusing mainly on your GR diagram, and how it is different from the ER diagram that was mapped.
2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.
2. 根据一个具体的小案例，编写并解释一条SQL查询语句，以提供该案例所需的答案。

Question 6 – Creating Databases with DDL

Tasks: 1. 提交你的作业4，并解释你是如何编写CREATE TABLE语句的，选择了哪些数据类型、键和约束，以及选择的原因。

1. Present your **Assignment 4** and explain how you wrote your CREATE TABLE statements, what types, keys, and constraints you chose, and why.
2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.
2. 根据一个具体的小案例，编写并解释一条SQL查询语句，以提供该案例所需的答案。

Question 7 – Functional Dependencies and Keys

Tasks:

1. 提交你的作业5，重点说明你发现的函数依赖关系以及它们如何帮助你选择主键。
1. Present your **Assignment 5**, focusing mainly on which functional dependencies you found, and how they helped you choose your keys.
2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.
2. 根据一个具体的小案例，编写并解释一条SQL查询语句，以提供该案例所需的答案。

Question 8 – Normalization to 1NF, 2NF, and 3NF

Tasks: 1. 提交你的作业5，重点说明你如何逐步将数据从非规范化形式规范化到第三范式(3NF)。

1. Present your **Assignment 5**, focusing mainly on how you normalized your data, step-by-step from unnormalized to 3NF.
2. Based on a small given case, write and explain an SQL query that provides the answer(s) needed in the case.
2. 根据一个具体的小案例，编写并解释一条SQL查询语句，以提供该案例所需的答案。